

RAG over EU Law

Ask anything about EU law. The answer is grounded in the acts in force, every claim is cited, and a citation opens the official PDF with the passage highlighted.

40,183

EU legal documents

all EU law in force, from EUR-Lex

841,805

Chunks

split by article and recital

RAGAS · 100 GOLDEN QUESTIONS

Faithfulness 0.98

Answer relevancy 0.85

Context recall 0.75

Context precision 0.69

FEATURES

pgvector HNSW

LangChain + Gemini

Voyage rerank-3

Structure-aware chunking

Redis exact + semantic cache

SSE streaming

PDF citations with highlights

RAGAS evaluation

LangSmith tracing

Live analytics

Rate limiting

Docker + Caddy

GitHub Actions CI/CD

DEVELOPMENT COST

€3.46

RAGAS evaluation after each change	€3.46
Oracle Cloud free tier · Ampere, 2 cores, 11 GB	€0
Voyage rerank-3 · 200M free tokens, used in evaluation	€0
Embedding all chunks · local model	€0

TRY IT LIVE

rag.elsayed2002.tech



Answers backed by the actual law.

The screenshot shows the RAG at Scale application interface. On the left, a sidebar contains a 'New chat' button and a search bar. The main chat area displays a conversation about flight compensation. The user's question is: 'What compensation do air passengers get when their flight is cancelled?'. The application's response, marked with a red 'R' icon, states that passengers have the right to compensation according to Article 7, unless specific exceptions apply. It lists the standard compensation amounts: EUR 250 for flights of 1,500 km or less, EUR 400 for intra-Community flights between 1,500 and 3,500 km, and EUR 600 for all other flights. The response also mentions that the operating air carrier may reduce this compensation by 50% if the passenger is offered re-routing. The sources listed at the bottom are 'Regulation (EC) No 261/2004' from page 4 and page 5. On the right, a preview of the source document is shown, highlighting the relevant articles from the Official Journal of the European Union.

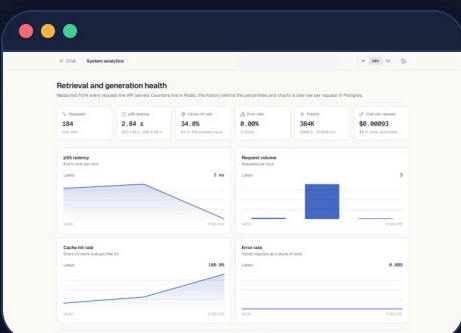
THE ANSWER SAYS

"EUR 250 for all flights of 1,500 kilometres or less ¹"



THE LAW SAYS

The screenshot shows the relevant part of the European Union Official Journal document, specifically Article 7 on the 'Right to compensation'. It lists three compensation levels: (a) EUR 250 for flights of 1,500 km or less, (b) EUR 400 for intra-Community flights between 1,500 and 3,500 km, and (c) EUR 600 for all other flights. It also mentions that the compensation may be reduced by 50% if the passenger is offered re-routing. The document is dated 17.2.2004 and is in English.



Live analytics

Every request is measured and stored. The dashboard shows:

p50 · p90 · p95 latency

Latency distribution

Time per step

Cache hit rate

Error rate

Tokens in / out

Cost per request

Requests per hour

Slowest queries

Repeated questions

1 h · 24 h · 7 d

Chunk by legal structure, not by size.

1

From PDFs to legal structure

Hit@6: right passage among the top 6 · same model, only the chunking differs

Version 1 · 43,106 PDFs

Recursive splitting, ~1,600 characters

200 overlap · 1,062,613 chunks

Hit@6 **32%**

on 2,968 test documents · 351,858 chunks



Version 2 · 40,183 HTML documents

One chunk per article and recital

841,805 chunks

Hit@6 **46%**

same 2,968 documents · 330,850 chunks

2

Every chunk knows where it is in the law

the Article 7 chunk behind the answer on page 2

Regulation (EC) No 261/2004 ... | Article 7
Right to compensation

1. Where reference is made to this Article, passengers shall receive compensation amounting to:

(a) EUR 250 for all flights of 1500 kilometres or less;

(b) EUR 400 for all intra-Community flights of more than 1500 kilometres, and for all other flights between 1500 and 3500 kilometres;

(c) EUR 600 for all flights not falling under (a) or (b). ...

Articles: one chunk each. Long ones are split at their numbered paragraphs (max 480 tokens).

Recitals: a few per chunk (max 250 tokens), labelled like "Recitals (14)-(15)".

Tables of numbers in annexes are skipped: they match no question.

841,805 vectors: embedded for free, searched in milliseconds.

1

Embedding

turning 841,805 chunks into vectors

Gemini · free tier

2.3 years

1,000 per day

Gemini · paid

~\$55

every re-embedding

bge-base · chosen

\$0 · 1 hr

free Kaggle GPU

Gemini is more accurate, but every new chunking means paying again. So I kept the free model and improved the other steps.

2

Vector database: building HNSW

PostgreSQL + pgvector · so a question isn't compared with every vector

Without an index

5.6 s

compares the question with every vector



With the HNSW index

12 ms

walks a graph to the nearest ones

460×

faster

99% the same results

Tuning the index



Built in 14 minutes

1.6 GB, kept in memory

16-bit vectors: half the size

Finding the right passage.

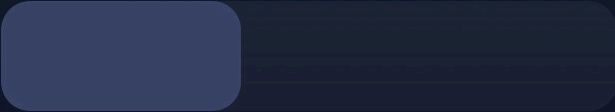
1

Reranking

Hit@6 over all 841,805 chunks: right passage among the 6 results the model reads

Vector search alone

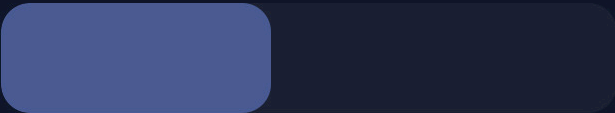
the 6 chunks closest in meaning



39%

+ at most 1 recital

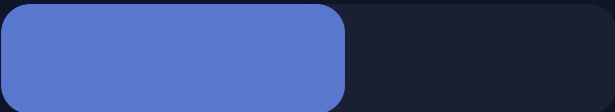
recitals explain why a law exists; they crowded out the rules



44%

+ local reranker, top 20

a small model re-reads the best 20 with the question



56%

+ Voyage rerank-3, top 100

a stronger model re-reads the best 100



80%

Vector search finds the right passage somewhere in its top 100 for **94%** of questions. The reranker moves it into the top 6 for **80%**.

2

How the system was measured

every change tested before it shipped

110 golden questions
with verified answers, 10
of them out of scope

Hit@6 after every
change: free, no LLM
needed

RAGAS with an LLM
judge for the final
answers

Running it for real.

1

Caching: fast answers, lower cost

measured live, from outside, over 170 requests

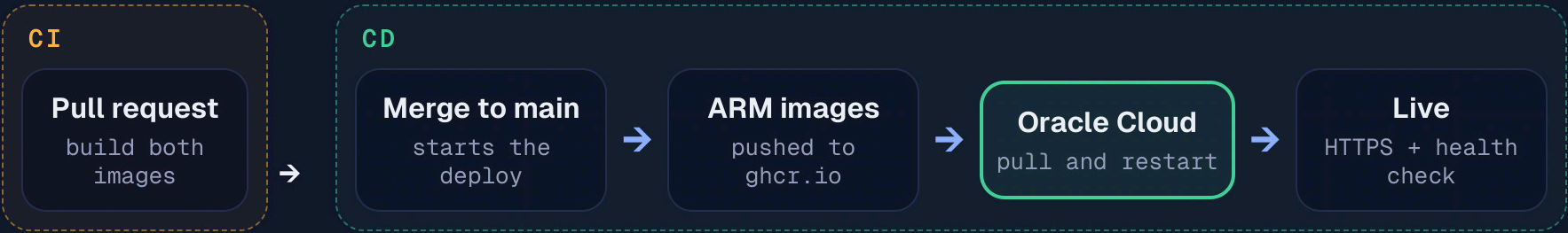
First words 1.4 s streamed as written	Full answer 2.1 s median	Same question 76 ms exact · \$0	Similar question 201 ms semantic · \$0
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A cached answer skips search, reranker and Gemini: it costs nothing. Similar questions are reused only at ≥ 0.95 similarity ("wage in 2026" vs "2025" is 0.93).

2

CI/CD pipeline

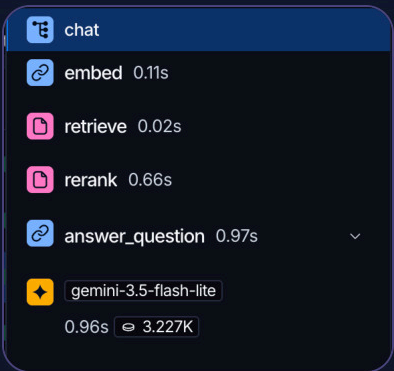
GitHub Actions · every merge to main goes live in ~5 minutes



3

Observability

every answer traced in LangSmith



Every step of every answer is logged. We can see each step's **input** and **output**.

- Every request logged
- Time per step

Try it live
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Source code
github.com/mostafaelsayed2002/rag-at-scale